

Theoretical Answers

Question 1: Medical Image with Dark Region Details

Answer: The logarithmic transformation is more suitable for revealing details in dark regions.

Reasoning:

- The log transform has the mathematical property of expanding dark pixel values while compressing bright ones.
- The derivative $d(s)/d(r) = 1/(1+r)$ is larger for small r values (dark regions), meaning more stretching occurs in the dark portion of the dynamic range.
- As $r \rightarrow 0$, the slope is steep (near 1), providing significant enhancement to barely visible dark details.
- In contrast, exponential transform does the opposite: it expands bright regions and compresses dark ones, making dark details even harder to distinguish.

Question 2: Overexposed Image Correction with Gamma

Answer: The gamma parameter (γ) should be greater than 1.

Reasoning:

- Gamma correction formula: $s = r^\gamma$ (with $c=1$)
- When $\gamma > 1$, the function is concave downward, mapping bright input values to darker output values.
- For example: with $\gamma = 2.2$, a pixel value $r = 0.8$ becomes $s = 0.64$, reducing brightness by 20%.
- This compresses the bright regions (where overexposure caused detail loss) while moderately affecting mid-tones.
- The transformation "pulls down" the washed-out highlights, restoring texture and detail that were clipped by the camera's overexposure.

